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**Problem 1.** ***CLT with Two-Way Dependence.*** *We follow the description of Menzel (2021) in Lecture note 17. Recall the representation of the RCE array $Z = (Z_{gh})_{1\le g,h<\infty}$:*

$$Z_{gh} = f(\alpha, \beta_g, \gamma_h, \varepsilon_{gh}), \qquad 1 \le g, h < \infty,$$

*where $\alpha$, $(\beta_g)_{g=1}^{\infty}$, $(\gamma_h)_{h=1}^{\infty}$, and $(\varepsilon_{gh})_{1\le g,h<\infty}$ are IID $U[0,1]$. We condition on $\alpha$ and drop it in what follows. We assume $Z_{gh}$ is scalar with mean zero and finite variance. Recall*

$$Z_{gh} = a_g + b_h + v_{gh} + e_{gh},$$

*where*

$$\begin{aligned} a_g &= \mathbb{E}[Z_{gh} \mid \beta_g], & v_{gh} &= \mathbb{E}[Z_{gh} \mid \beta_g, \gamma_h] - \mathbb{E}[Z_{gh} \mid \beta_g] - \mathbb{E}[Z_{gh} \mid \gamma_h], \\ b_h &= \mathbb{E}[Z_{gh} \mid \gamma_h], & e_{gh} &= Z_{gh} - \mathbb{E}[Z_{gh} \mid \beta_g, \gamma_h]. \end{aligned}$$

*Assume as in note 17 that*

$$v_{gh} = c_g d_h, \qquad c_g = g(\beta_g), \qquad d_h = d(\gamma_h).$$

*Suppose we observe $(Z_{gh})$ for $g = 1, \dots, G$ and $h = 1, \dots, G$. Define the $5 \times 1$ vector*

$$S_G = \begin{pmatrix} \sqrt{G}\,\bar{a}_G \\ \sqrt{G}\,\bar{b}_G \\ \sqrt{G}\,\bar{c}_G \\ \sqrt{G}\,\bar{d}_G \\ G\,\bar{e}_{GG} \end{pmatrix},$$

*where $\bar{a}_G = \frac{1}{G}\sum_{g=1}^{G} a_g$, $\bar{b}_G$, $\bar{c}_G$, and $\bar{d}_G$ are defined analogously, and $\bar{e}_{GG} = \frac{1}{GG}\sum_{g=1}^{G}\sum_{h=1}^{G} e_{gh}$. You will show that*

$$S_G \to_d N(0, \Sigma)$$

*as $G \to \infty$, where the expression for $\Sigma$ is given in note 17.*

*Consider the $\sigma$-algebras*

$$\mathcal{F}_{G,g} = \begin{cases} \{\Omega, \emptyset\}, & g = 0, \\ \sigma\big(\beta_1, \dots, \beta_g,\ \gamma_1, \dots, \gamma_g,\ (\varepsilon_{g'1})_{g'=1}^{G},\ \dots,\ (\varepsilon_{g'g})_{g'=1}^{G}\big), & g = 1, \dots, G. \end{cases}$$

*(a) Calculate $\mathbb{E}[S_G \mid \mathcal{F}_{G,g}]$ for $g = 0, \dots, G$. Give an expression for the martingale differences $X_{G,g} = \mathbb{E}[S_G \mid \mathcal{F}_{G,g}] - \mathbb{E}[S_G \mid \mathcal{F}_{G,g-1}]$, $g = 1, \dots, G$.*

*(b) Show that*

$$\max_{1\le g\le G} \|X_{G,g}\| \to_p 0$$

*as $G \to \infty$. State any additional assumptions you use.*

*(c) Show that*

$$\sum_{g=1}^{G} (X_{G,g})(X_{G,g})' \to_p \Sigma$$

*as $G \to \infty$. State any additional assumptions you use.*

*(d) Show that*

$$\limsup_{G\to\infty} \mathbb{E}\left[\max_{1\le g\le G} \|X_{G,g}\|^2\right] < \infty.$$

*State any additional assumptions you use.*

*In view of (b)–(d), it follows by a CLT for triangular arrays of martingale differences (e.g., Hall and Heyde, (1980, Theorem 3.2)) that $S_G \to_d N(0,\Sigma)$ as $G\to\infty$.*

**Solution.** *Preliminaries.* Under the Aldous–Hoover representation, the random variables $\{\beta_g\}_{g\ge1}$, $\{\gamma_h\}_{h\ge1}$, $\{\varepsilon_{gh}\}_{g,h\ge1}$ are mutually independent; $a_g$ and $c_g$ are $\sigma(\beta_g)$-measurable with mean zero, $b_h$ and $d_h$ are $\sigma(\gamma_h)$-measurable with mean zero, and the cell residual $e_{g'h}$ satisfies $\mathbb{E}[e_{g'h} \mid \beta_{g'}, \gamma_h] = 0$. Write $\nu(\beta,\gamma) := \mathbb{E}[Z_{11} \mid \beta_1 = \beta, \gamma_1 = \gamma]$ for the $(\beta,\gamma)$-conditional mean, so $e_{g'h} = Z_{g'h} - \nu(\beta_{g'}, \gamma_h)$. Define the $\beta$-marginalized residual

$$\tilde{e}(\gamma,\varepsilon) := \mathbb{E}[Z_{11} \mid \gamma_1 = \gamma, \varepsilon_{11} = \varepsilon] - b(\gamma),$$

so $\mathbb{E}[\tilde{e}(\gamma_h, \varepsilon_{g'h}) \mid \gamma_h] = 0$, $\mathbb{E}[\tilde{e}(\gamma_h,\varepsilon_{g'h})^2] < \infty$ (inherited from $\mathbb{E}[e^2] < \infty$). By construction $e_{g'h} - \tilde{e}(\gamma_h,\varepsilon_{g'h})$ has mean zero given $(\gamma_h, \varepsilon_{g'h})$, equivalently $\mathbb{E}[e_{g'h} \mid \gamma_h, \varepsilon_{g'h}] = \tilde{e}(\gamma_h,\varepsilon_{g'h})$.

(a) *Components 1–4.* Since $a_\ell$ is $\sigma(\beta_\ell)$-measurable, $\mathbb{E}[a_\ell \mid \mathcal{F}_{G,g}] = a_\ell \mathbf{1}\{\ell \le g\}$ (and equals $\mathbb{E}[a_\ell] = 0$ when $\ell > g$ by independence). Hence

$$\mathbb{E}[\sqrt{G}\,\bar{a}_G \mid \mathcal{F}_{G,g}] = \frac{1}{\sqrt{G}} \sum_{\ell=1}^{g} a_\ell,$$

and analogously for the other three components, replacing $a_\ell$ by $b_\ell$, $c_\ell$, and $d_\ell$ respectively:

$$\mathbb{E}[\sqrt{G}\,\bar{b}_G \mid \mathcal{F}_{G,g}] = \tfrac{1}{\sqrt{G}} \sum_{\ell=1}^{g} b_\ell, \qquad \mathbb{E}[\sqrt{G}\,\bar{c}_G \mid \mathcal{F}_{G,g}] = \tfrac{1}{\sqrt{G}} \sum_{\ell=1}^{g} c_\ell,$$

$$\mathbb{E}[\sqrt{G}\,\bar{d}_G \mid \mathcal{F}_{G,g}] = \tfrac{1}{\sqrt{G}} \sum_{\ell=1}^{g} d_\ell.$$

*Component 5.* Compute $\mathbb{E}[e_{g'h} \mid \mathcal{F}_{G,g}]$ by casework on $(g', h)$:

- *(i) $g' \le g$, $h \le g$:* all of $\beta_{g'}, \gamma_h, \varepsilon_{g'h}$ lie in $\mathcal{F}_{G,g}$, so $e_{g'h}$ itself is $\mathcal{F}_{G,g}$-measurable and $\mathbb{E}[e_{g'h} \mid \mathcal{F}_{G,g}] = e_{g'h}$.
- *(ii) $g' \le g$, $h > g$:* $\beta_{g'}$ is in $\mathcal{F}_{G,g}$; $\gamma_h$ and $\varepsilon_{g'h}$ are independent of $\mathcal{F}_{G,g}$. So $\mathbb{E}[Z_{g'h} \mid \mathcal{F}_{G,g}] = \mathbb{E}[Z_{g'h} \mid \beta_{g'}] = a_{g'}$ and $\mathbb{E}[\nu(\beta_{g'},\gamma_h) \mid \mathcal{F}_{G,g}] = \mathbb{E}[\nu(\beta_{g'},\gamma_h) \mid \beta_{g'}] = a_{g'}$, so $\mathbb{E}[e_{g'h} \mid \mathcal{F}_{G,g}] = 0$.
- *(iii) $g' > g$, $h \le g$:* $\gamma_h$ and $\varepsilon_{g'h}$ are in $\mathcal{F}_{G,g}$; $\beta_{g'}$ is independent of $\mathcal{F}_{G,g}$. Hence $\mathbb{E}[Z_{g'h} \mid \mathcal{F}_{G,g}] = \mathbb{E}[Z_{g'h} \mid \gamma_h, \varepsilon_{g'h}] = b_h + \tilde{e}(\gamma_h,\varepsilon_{g'h})$ and $\mathbb{E}[\nu(\beta_{g'},\gamma_h) \mid \mathcal{F}_{G,g}] = b_h$, so $\mathbb{E}[e_{g'h} \mid \mathcal{F}_{G,g}] = \tilde{e}(\gamma_h,\varepsilon_{g'h})$.
- *(iv) $g' > g$, $h > g$:* all latent coordinates are independent of $\mathcal{F}_{G,g}$, so $\mathbb{E}[Z_{g'h} \mid \mathcal{F}_{G,g}] = 0$ and $\mathbb{E}[\nu(\beta_{g'},\gamma_h) \mid \mathcal{F}_{G,g}] = 0$, giving $\mathbb{E}[e_{g'h} \mid \mathcal{F}_{G,g}] = 0$.

Summing,

$$\mathbb{E}[G\,\bar{e}_{GG} \mid \mathcal{F}_{G,g}] = \tfrac{1}{G}\sum_{g'\le g}\sum_{h\le g} e_{g'h} \;+\; \tfrac{1}{G}\sum_{g'>g}\sum_{h\le g}\tilde{e}(\gamma_h,\varepsilon_{g'h}).$$

*Martingale differences.* Collecting,

$$X_{G,g} = \tfrac{1}{\sqrt{G}}(a_g,\, b_g,\, c_g,\, d_g,\, 0)' + (0,0,0,0,\Delta_g)',$$

where taking the difference of $\mathbb{E}[G\,\bar{e}_{GG} \mid \mathcal{F}_{G,g}]$ at steps $g$ and $g-1$ yields

$$\Delta_g = \tfrac{1}{G}\Bigg[\underbrace{\sum_{g'=1}^{g} e_{g'g}}_{\text{column } g\text{, fully revealed}} + \underbrace{\sum_{g'=g+1}^{G}\tilde{e}(\gamma_g,\varepsilon_{g'g})}_{\text{column } g\text{, projected}} + \underbrace{\sum_{h=1}^{g-1}\big(e_{gh}-\tilde{e}(\gamma_h,\varepsilon_{gh})\big)}_{\text{row } g\text{ cells, projection upgraded}}\Bigg].$$

The three sums correspond to the cells whose status changes at step $g$: column $g$ cells become fully measurable (Case (i)) or move from Case (iv) to Case (iii); row $g$ cells with $h<g$ move from Case (iii) to Case (i), replacing $\tilde{e}(\gamma_h,\varepsilon_{gh})$ by $e_{gh}$. Straightforward verification using $\mathbb{E}[a_g]=\mathbb{E}[b_g]=\mathbb{E}[c_g]=\mathbb{E}[d_g]=0$ and $\mathbb{E}[e_{gh}-\tilde{e}(\gamma_h,\varepsilon_{gh}) \mid \gamma_h,\varepsilon_{gh}]=0$ gives $\mathbb{E}[X_{G,g} \mid \mathcal{F}_{G,g-1}]=0$.

(b) For components 1–4, fix $\epsilon>0$:

$$\mathbb{P}\Big(\max_{g\le G}|a_g|>\epsilon\sqrt{G}\Big) \le G\,\mathbb{P}(|a_1|>\epsilon\sqrt{G}) \le \frac{\mathbb{E}\big[a_1^2\,\mathbf{1}\{|a_1|>\epsilon\sqrt{G}\}\big]}{\epsilon^2},$$

using $\mathbb{P}(|a_1|>\epsilon\sqrt{G}) \le \mathbb{E}[a_1^2\mathbf{1}\{|a_1|>\epsilon\sqrt{G}\}]/(\epsilon^2 G)$. The right-hand side vanishes by dominated convergence as $\mathbf{1}\{|a_1|>\epsilon\sqrt{G}\}\to 0$ a.s. and $a_1^2$ is integrable. The same argument applies to $b_g, c_g, d_g$.

For component 5, write $\Delta_g = G^{-1}(S_{1,g}+S_{2,g}+S_{3,g})$ with

$$S_{1,g}=\sum_{g'=1}^{g} e_{g'g}, \qquad S_{2,g}=\sum_{g'=g+1}^{G}\tilde{e}(\gamma_g,\varepsilon_{g'g}), \qquad S_{3,g}=\sum_{h=1}^{g-1}\big(e_{gh}-\tilde{e}(\gamma_h,\varepsilon_{gh})\big).$$

*Conditional independence.* Conditional on $\gamma_g$, $(\beta_{g'},\varepsilon_{g'g})_{g'=1,\dots,g}$ are iid and jointly independent of $\gamma_g$ (by mutual independence of Aldous–Hoover latents); hence $\{e_{g'g}\}_{g'=1}^{g}$ are conditionally iid given $\gamma_g$ with $\mathbb{E}[e_{g'g} \mid \gamma_g] = \mathbb{E}[\mathbb{E}[e_{g'g} \mid \beta_{g'},\gamma_g] \mid \gamma_g] = 0$. Identical logic applies to $S_{2,g}$ (summands $\tilde{e}(\gamma_g,\varepsilon_{g'g})$ conditionally iid across $g'>g$ given $\gamma_g$) and to $S_{3,g}$ (summands $e_{gh}-\tilde{e}(\gamma_h,\varepsilon_{gh})$ conditionally iid across $h<g$ given $\beta_g$, with conditional mean zero by construction of $\tilde{e}$).

*Rosenthal bound.* For any $p\in(2,2+\delta]$,

$$\mathbb{E}\big[|S_{1,g}|^p \mid \gamma_g\big] \le C_p\big\{g^{p/2}\big(\mathbb{E}[e_{11}^2 \mid \gamma_g]\big)^{p/2} + g\,\mathbb{E}[|e_{11}|^p \mid \gamma_g]\big\} = O_{a.s.}(g^{p/2}).$$

Taking outer expectation, $\mathbb{E}|S_{1,g}|^p \le C\,g^{p/2}$ (uniformly in $g\le G$). Analogously, $\mathbb{E}|S_{2,g}|^p \le C\,(G-g)^{p/2}$ and $\mathbb{E}|S_{3,g}|^p \le C\,g^{p/2}$. Since $\max(g,G-g)^{p/2} \le G^{p/2}$,

$$\mathbb{E}|\Delta_g|^p \le G^{-p}\cdot 3^{p-1}\big(\mathbb{E}|S_{1,g}|^p + \mathbb{E}|S_{2,g}|^p + \mathbb{E}|S_{3,g}|^p\big) \le C'\,G^{-p/2}$$

uniformly in $1 \le g \le G$. Therefore

$$\mathbb{P}\Big(\max_{g\le G}|\Delta_g| > \epsilon\Big) \le G \cdot \frac{C'\,G^{-p/2}}{\epsilon^p} = \frac{O(G^{1-p/2})}{\epsilon^p} \to 0$$

for any $p > 2$. Combining, $\max_g \|X_{G,g}\| \to_p 0$.

*Assumptions used:* $\mathbb{E}[a_1^2], \mathbb{E}[b_1^2], \mathbb{E}[c_1^2], \mathbb{E}[d_1^2] < \infty$ (for components 1–4, via dominated convergence); $\mathbb{E}|e_{11}|^{2+\delta} < \infty$ for some $\delta > 0$ (for component 5, via Rosenthal).

(c) Decompose $\sum_g X_{G,g}X_{G,g}' = A_G + B_G + C_G$, where

$$A_G = \tfrac{1}{G}\sum_{g=1}^{G} w_g w_g' \text{ with } w_g = (a_g, b_g, c_g, d_g)', \text{ extended by a zero } 5^{\text{th}} \text{ row},$$

$$B_G = \sum_{g=1}^{G} \Delta_g^2 \text{ in the } (5,5) \text{ entry, zero elsewhere},$$

$$C_G = \sum_{g=1}^{G} \tfrac{1}{\sqrt{G}}(a_g, b_g, c_g, d_g)'\,\Delta_g \quad \text{(rank-one cross-terms)}.$$

$A_G \to_p \mathbb{E}[w_1 w_1']$ by the LLN for iid sequences with finite second moments. The limit is a $4\times 4$ block $\Sigma_w$ whose entries are derived as follows: $a_1, c_1$ are $\sigma(\beta_1)$-measurable and $b_1, d_1$ are $\sigma(\gamma_1)$-measurable, with $\beta_1 \perp\!\!\!\perp \gamma_1$ and all four marginals mean zero. Hence

$$\mathbb{E}[a_1 b_1] = \mathbb{E}[a_1]\mathbb{E}[b_1] = 0, \qquad \mathbb{E}[a_1 d_1] = \mathbb{E}[b_1 c_1] = \mathbb{E}[c_1 d_1] = 0,$$

while $\mathbb{E}[a_1 c_1]$ and $\mathbb{E}[b_1 d_1]$ are generally nonzero (both terms are functions of the same latent). Diagonals are $\sigma_a^2, \sigma_b^2, \sigma_c^2, \sigma_d^2$, giving the block $\Sigma_w$ with two non-zero off-diagonal pairs $\mathbb{E}[a_1 c_1]$ and $\mathbb{E}[b_1 d_1]$, in agreement with Lecture 17.

*$B_G$ convergence and cross-g orthogonality.* The claim is $B_G \to_p \sigma_e^2 := \mathbb{E}[e_{11}^2]$. Recall $\Delta_g = G^{-1}(S_{1,g}+S_{2,g}+S_{3,g})$ from (b). The conditional expectations of step (a) telescope across $g = 0, \ldots, G$ to $\mathbb{E}[G\bar{e}_{GG} \mid \mathcal{F}_{G,G}] - \mathbb{E}[G\bar{e}_{GG} \mid \mathcal{F}_{G,0}] = G\bar{e}_{GG}$, so $\sum_{g=1}^{G}\Delta_g = G\bar{e}_{GG}$. Since $\{\Delta_g\}_{g=1}^{G}$ are martingale differences (by the MD property of $X_{G,g}$), MD orthogonality gives $\sum_g \mathbb{E}[\Delta_g^2] = \mathrm{Var}\big(\sum_g \Delta_g\big) = \mathrm{Var}(G\bar{e}_{GG})$:

$$\mathrm{Var}(G\bar{e}_{GG}) = G^{-2}\sum_{g',h,g'',h'} \mathrm{Cov}(e_{g'h}, e_{g''h'}) = G^{-2}\sum_{g',h}\mathbb{E}[e_{g'h}^2] = \mathbb{E}[e_{11}^2] = \sigma_e^2,$$

using pairwise orthogonality of distinct cell residuals: for $(g',h) \neq (g'',h')$, $\mathbb{E}[e_{g'h}e_{g''h'}] = 0$ by $\mathbb{E}[e \mid \beta, \gamma] = 0$ (condition on the full $(\beta,\gamma)$-vector generating both cells and factor the expectation). For convergence in probability, Chebyshev plus a second-moment bound on $B_G - \mathbb{E}[B_G]$ suffices: by the Rosenthal bound from (b), $\mathrm{Var}(\Delta_g^2) = O(G^{-2})$, so $\mathrm{Var}(B_G) \le G \cdot O(G^{-2}) = O(G^{-1}) \to 0$.

*Cross-term $C_G$.* The claim is $C_G \to_p 0$. For each $g$, $(w_g, \Delta_g)$ satisfy

$$\mathbb{E}[w_g\Delta_g \mid \mathcal{F}_{G,g-1}] = w_g \cdot \mathbb{E}[\Delta_g \mid \mathcal{F}_{G,g-1}] = 0$$

because $w_g = (a_g, b_g, c_g, d_g)'$ is $\sigma(\beta_g, \gamma_g)$-measurable hence independent of $\mathcal{F}_{G,g-1}$, and $\Delta_g$ has mean zero conditional on $\mathcal{F}_{G,g-1}$; joint conditional mean zero follows from the tower property combined with independence of $w_g$ from the cells entering $\Delta_g$ given $\mathcal{F}_{G,g-1}$. Hence $\{w_g\Delta_g/\sqrt{G}\}_{g=1}^G$ is a martingale-difference sequence, and by orthogonality

$$\mathbb{E}\Big\|\sum_g \tfrac{w_g\Delta_g}{\sqrt{G}}\Big\|^2 = \tfrac{1}{G}\sum_g \mathbb{E}[\|w_g\|^2\Delta_g^2] \le \tfrac{1}{G}\sum_g \big(\mathbb{E}\|w_g\|^4\big)^{1/2}\big(\mathbb{E}\Delta_g^4\big)^{1/2}.$$

Under $\mathbb{E}\|w_1\|^4 < \infty$ and the Rosenthal bound $\mathbb{E}\Delta_g^4 = O(G^{-2})$ (taking $p = 4$ in (b), requiring $\delta \ge 2$, i.e. $\mathbb{E}|e_{11}|^4 < \infty$), the sum is $O(G^{-1}) \to 0$, so $C_G \to_p 0$ by Markov.

Assembling: $\sum_g X_{G,g}X_{G,g}' \to_p \Sigma$ where $\Sigma$ has the $4\times 4$ block $\Sigma_w$ in its upper-left $4\times 4$ block and $\sigma_e^2$ in its $(5,5)$ entry, all other entries zero — precisely the $\Sigma$ of Lecture 17.

*Assumptions used:* $\mathbb{E}\|w_1\|^4 < \infty$, i.e. $\mathbb{E}|a_1|^4, \mathbb{E}|b_1|^4, \mathbb{E}|c_1|^4, \mathbb{E}|d_1|^4 < \infty$; $\mathbb{E}|e_{11}|^4 < \infty$ (for the Rosenthal bound with $p = 4$). These are implied by $\mathbb{E}|Z_{11}|^4 < \infty$.

(d) For components 1–4, using $\max_g a_g^2 \le \sum_g a_g^2$,

$$\mathbb{E}\Big[\max_{g\le G}\tfrac{a_g^2}{G}\Big] \le \tfrac{1}{G}\sum_{g=1}^G \mathbb{E}[a_g^2] = \mathbb{E}[a_1^2] < \infty,$$

uniformly in $G$, and likewise for $b_g^2, c_g^2, d_g^2$. For component 5, by Jensen's inequality and the Rosenthal bound from (b) with $p = 2+\delta$:

$$\begin{aligned}\mathbb{E}\Big[\max_{g\le G}\Delta_g^2\Big] &\le \Big(\mathbb{E}\Big[\max_{g\le G}|\Delta_g|^{2+\delta}\Big]\Big)^{2/(2+\delta)} \le \Big(\sum_{g=1}^G \mathbb{E}|\Delta_g|^{2+\delta}\Big)^{2/(2+\delta)} \\ &\le \big(G\cdot O(G^{-(2+\delta)/2})\big)^{2/(2+\delta)} = O(G^{-\delta/(2+\delta)}),\end{aligned}$$

which is bounded (indeed vanishing). Summing the five component bounds gives $\limsup_{G\to\infty}\mathbb{E}[\max_g\|X_{G,g}\|^2] \le \mathbb{E}[a_1^2] + \mathbb{E}[b_1^2] + \mathbb{E}[c_1^2] + \mathbb{E}[d_1^2] + 0 < \infty$.

*Assumptions used:* finite second moments for $a, b, c, d$; $\mathbb{E}|e_{11}|^{2+\delta} < \infty$ for some $\delta > 0$ (for the max-over-$g$ bound on $\Delta_g^2$ via the Rosenthal argument of (b)).

Parts (b)–(d) verify the three conditions of the martingale CLT for triangular arrays of Hall and Heyde (1980, Theorem 3.2) — the martingale-difference version of the stable-convergence CLT stated as Theorem 1 of Lecture 10 (cf. Remark 1 therein, extending to triangular arrays). Hence $S_G \to_d N(0,\Sigma)$ as $G \to \infty$. □

**Problem 2.** ***Narrative identification.*** *Suppose you wish to estimate the parameters $A$ and $B$ in the model*

$$Y_t'A = X_t'B + u_t',$$

*where $Y_t$ is $m\times 1$, $A$ is $m\times m$ and invertible, $X_t$ is $p\times 1$, $B$ is $p\times m$, and $u_t$ is $m\times 1$, where $p \ge m$. You observe an IID sample $(X_t, Y_t)$ for $t = 1,\dots,n$.*[1] *We treat $m$ as fixed and*

[1] Typically this is done in an SVAR context so there is time-series dependence, but that's the least of our worries.

*let $n \to \infty$. The errors $u_t$ are assumed to satisfy $\mathbb{E}[X_t u_t'] = 0$ and $\mathbb{E}[u_t u_t'] = I$, the $m \times m$ identity matrix. You may also assume $\mathbb{E}[X_t X_t']$ is finite and positive definite.*

*Rewrite the model as*

$$Y_t' = X_t' B A^{-1} + u_t' A^{-1} = X_t' C + v_t',$$

*where $C = BA^{-1}$ and $v_t = (A^{-1})' u_t$ has variance $\Sigma = (A^{-1})' A^{-1}$. Evidently, there is an identification problem: we can identify $C$ from the moment $\mathbb{E}[X_t Y_t']$ and $\Sigma$ from the moment $\mathbb{E}[v_t v_t']$, but we cannot disentangle $A$ and $B$ from $C$ and $\Sigma$. Let $h(\Sigma)$ denote the lower-triangular Cholesky factor of $\Sigma$. That is, the unique lower-triangular matrix with positive diagonal elements for which $h(\Sigma)h(\Sigma)' = \Sigma$. For any rotation matrix $Q$ (i.e., a square matrix for which $QQ' = Q'Q = I$), we also have $h(\Sigma)Q(h(\Sigma)Q)' = h(\Sigma)QQ'h(\Sigma)' = \Sigma$. Letting $\mathcal{Q}$ denote the set of all such $Q$, the identified set for $A$ is $\{(h(\Sigma)')^{-1}Q : Q \in \mathcal{Q}\}$ and the identified set for $B$ is $\{C(h(\Sigma)')^{-1}Q : Q \in \mathcal{Q}\}$.*

*Several authors (see, e.g., Antolín-Díaz and Rubio-Ramírez (*AER*, 2018); Ludvigson, Ma, and Ng (*AEJ:Macro*, 2021); Giacomini, Kitagawa, and Read (WP, 2025)) have advocated using "narrative restrictions". In effect, the restrictions constrain the realizations of $u_t$ on certain dates, which shrinks the identified set of $A$ and $B$. Note that $u_t' = v_t' A = v_t'(h(\Sigma)')^{-1}Q$ for some $Q \in \mathcal{Q}$. Restricting $u_t'$ on certain dates $t$ allows us, with enough restrictions, to pin down a unique $Q \in \mathcal{Q}$, and thus "identify" the parameters $A$ and $B$. For simplicity, let us assume that it is known that*

$$R = \begin{pmatrix} u_1' \\ \vdots \\ u_m' \end{pmatrix},$$

*where $R$ is $m \times m$. Then*

$$R = \underbrace{\begin{pmatrix} v_1' \\ \vdots \\ v_m' \end{pmatrix}}_{=V} (h(\Sigma)')^{-1} Q,$$

*and so, assuming invertibility of $V$, we have*

$$Q = h(\Sigma)' V^{-1} R.$$

*Suppose you estimate $C$ by regressing $Y_t$ on $X_t$:*

$$\hat{C} = \left( \frac{1}{n} \sum_{t=1}^{n} X_t X_t' \right)^{-1} \left( \frac{1}{n} \sum_{t=1}^{n} X_t Y_t' \right),$$

*and estimate $\Sigma$ from the OLS residuals:*

$$\hat{\Sigma} = \frac{1}{n} \sum_{t=1}^{n} \hat{v}_t \hat{v}_t',$$

*where* $\hat{v}_t' = Y_t' - X_t'\hat{C}$. *For any* $Q \in \mathcal{Q}$, *you can then estimate* $B$ *using*

$$\hat{C}(h(\hat{\Sigma})')^{-1}Q.$$

*For the specific* $Q$ *corresponding to the narrative restriction,*

$$\begin{aligned}\hat{B} &= \hat{C}(h(\hat{\Sigma})')^{-1}h(\hat{\Sigma})'\hat{V}^{-1}R \\ &= \hat{C}\hat{V}^{-1}R,\end{aligned}$$

*where*

$$\hat{V} = \begin{pmatrix} \hat{v}_1' \\ \vdots \\ \hat{v}_m' \end{pmatrix}.$$

*Derive the asymptotic distribution of* $\hat{B}$ *and use this, if possible, to develop a suitable inference procedure. State any additional assumptions you use.*

**Solution.** Write $\Delta C = \hat{C} - C$ $(p \times m)$ and $\Delta V = \hat{V} - V$ $(m \times m)$. Throughout this problem, $\Sigma$ denotes the $m \times m$ reduced-form variance $\mathbb{E}[v_t v_t'] = (A^{-1})'A^{-1}$ (unrelated to the $5 \times 5$ symbol $\Sigma$ of Problem 1). The argument proceeds in six steps.

1. *Asymptotic expansion of* $\hat{C}$. Under iid sampling with $\mathbb{E}\|X_t\|^{2+\delta} < \infty$, $\mathbb{E}\|u_t\|^{2+\delta} < \infty$ for some $\delta > 0$, conditional homoskedasticity $\mathbb{E}[v_t v_t' \mid X_t] = \Sigma$, and $Q_{xx} := \mathbb{E}[X_t X_t']$ invertible, the standard OLS CLT yields

$$\sqrt{n}\,\mathrm{vec}(\Delta C) \to_d \xi \sim N(0, \Omega_C), \qquad \Omega_C = \Sigma \otimes Q_{xx}^{-1},$$

where $\Sigma = \mathbb{E}[v_t v_t']$. This follows from $\mathrm{vec}(\Delta C) = (I_m \otimes (X'X/n)^{-1}) \cdot n^{-1/2} \sum_t \mathrm{vec}(X_t v_t')$ with $\mathrm{vec}(X_t v_t') = v_t \otimes X_t$, and Lindeberg–Lévy applied to the iid zero-mean summands $v_t \otimes X_t$ under the stated moment conditions ($\mathrm{Var}(v_t \otimes X_t) = \Sigma \otimes Q_{xx}$).

2. $\Delta V$ *in terms of* $\Delta C$. $\hat{v}_t = Y_t - \hat{C}'X_t = v_t - (\hat{C}' - C')X_t$, so $\hat{v}_t - v_t = -\Delta C' X_t$ and, stacking $t = 1, \ldots, m$,

$$\hat{V} - V = -\tilde{X}'\Delta C, \qquad \tilde{X} := [X_1 \mid X_2 \mid \cdots \mid X_m] \in \mathbb{R}^{p \times m}.$$

3. *Asymptotic independence.* Decompose

$$n^{-1/2}\sum_{t=1}^{n} X_t v_t' = n^{-1/2}\sum_{t=1}^{m} X_t v_t' + n^{-1/2}\sum_{t=m+1}^{n} X_t v_t'.$$

The first piece is $O_p(1/\sqrt{n}) = o_p(1)$ since $m$ is fixed, and the second piece is independent of $(X_1, v_1, \ldots, X_m, v_m)$. Hence $\sqrt{n}\,\mathrm{vec}(\Delta C)$ is asymptotically independent of $\mathcal{G} := \sigma(X_1, v_1, \ldots, X_m, v_m)$; in particular, the limiting $\xi$ in step 1 can be taken independent of $\mathcal{G}$.

4. *Linearization of $\hat{B}$.* Since $V$ is invertible by assumption and $\hat{V} \to_p V$ (immediate from $\Delta V = -\tilde{X}'\Delta C = O_p(1/\sqrt{n})$), the smallest singular value satisfies $\sigma_{\min}(\hat{V}) \geq \sigma_{\min}(V) - \|\Delta V\| \geq \sigma_{\min}(V)/2 > 0$ on the event $\{\|\Delta V\| \leq \sigma_{\min}(V)/2\}$, whose probability tends to one. On this event, $\hat{V}$ is invertible and $\hat{V}^{-1} = V^{-1} - V^{-1}(\Delta V)V^{-1} + O_p(\|\Delta V\|^2) = V^{-1} - V^{-1}(\Delta V)V^{-1} + o_p(1/\sqrt{n})$. Thus,

$$\begin{aligned}\hat{B} - B &= (C + \Delta C)\big(V^{-1} - V^{-1}(\Delta V)V^{-1}\big)R - CV^{-1}R \\ &= (\Delta C)V^{-1}R - CV^{-1}(\Delta V)V^{-1}R + o_p(1/\sqrt{n}).\end{aligned}$$

   Substituting $\Delta V = -\tilde{X}'\Delta C$ from step 2,

$$\hat{B} - B = (\Delta C)\, V^{-1}R + CV^{-1}\tilde{X}'(\Delta C)V^{-1}R + o_p(1/\sqrt{n}) = M_n \cdot \Delta C \cdot N + o_p(1/\sqrt{n}),$$

   where $M_n := I_p + CV^{-1}\tilde{X}'$ (a $p \times p$ random matrix, $\mathcal{G}$-measurable) and $N := V^{-1}R$ (an $m \times m$ $\mathcal{G}$-measurable matrix).

5. *Vectorized limit and stable convergence.* Using $\operatorname{vec}(M_n(\Delta C)N) = (N' \otimes M_n)\operatorname{vec}(\Delta C)$,

$$\sqrt{n}\operatorname{vec}(\hat{B} - B) = K_n \cdot \sqrt{n}\operatorname{vec}(\Delta C) + o_p(1), \qquad K_n := N' \otimes M_n.$$

   Since $K_n$ is $\mathcal{G}$-measurable and $\sqrt{n}\operatorname{vec}(\Delta C) \to_d \xi$ $\mathcal{G}$-stably by step 3 (with $\xi$ independent of $\mathcal{G}$), the continuous-mapping/Slutsky variant for stable convergence (Lecture 10; see also the argument in PS3 Problem 1) gives

$$\sqrt{n}\operatorname{vec}(\hat{B} - B) \to_d K_n\,\xi \quad (\mathcal{G}\text{-stably}).$$

   The limit is a *mixed normal*: conditional on $\mathcal{G}$, $N(0, \Omega_B)$ with $\Omega_B := K_n\Omega_C K_n'$; unconditionally, a $\mathcal{G}$-mixture.

   *Structural simplification.* The conditional variance collapses cleanly. With $\Omega_C = \Sigma \otimes Q_{xx}^{-1}$ and $K_n = N' \otimes M_n$ (where $N = V^{-1}R$, $M_n = I_p + CV^{-1}\tilde{X}'$, both $\mathcal{G}$-measurable),

$$\Omega_B = (N' \otimes M_n)(\Sigma \otimes Q_{xx}^{-1})(N \otimes M_n') = (N'\Sigma N) \otimes (M_n Q_{xx}^{-1} M_n').$$

   The narrative-identification formula $V^{-1}R = A$ gives $N = A$. The structural normalization $\mathbb{E}[u_t u_t'] = I_m$ implies $\Sigma = (A^{-1})'(A^{-1}) = (AA')^{-1}$, hence

$$N'\Sigma N = A'(AA')^{-1}A = I_m.$$

   Therefore

$$\Omega_B = I_m \otimes (M_n\, Q_{xx}^{-1}\, M_n').$$

   The asymptotic variance is block-diagonal with $m$ identical $p \times p$ blocks: each column of $\hat{B}$ has covariance $M_n Q_{xx}^{-1} M_n'/n$, and columns are asymptotically uncorrelated. Notably, $\Omega_B$ does not depend on the unobserved $\Sigma$ or on $A$ directly — a feature that makes inference particularly clean.

6. *Inference.* Plug in $\hat{M}_n := I_p + \hat{C}\hat{V}^{-1}\tilde{X}'$ and $\hat{Q}_{xx} := n^{-1}\sum_t X_t X_t'$:

$$\hat{\Omega}_B := I_m \otimes (\hat{M}_n\, \hat{Q}_{xx}^{-1}\, \hat{M}_n').$$

*Consistency.* $\hat{C} \to_p C$ and $\hat{V} \to_p V$ from steps 1–2 imply $\hat{M}_n \to_p M_n$; $\hat{Q}_{xx} \to_p Q_{xx}$ by the LLN. Hence $\hat{\Omega}_B \to_p \Omega_B$.

*Multivariate t-statistic.* For any non-random contrast $a \in \mathbb{R}^{pm}$, applying the continuous mapping theorem for stable convergence (to the smooth map $g(K, \xi, W) = a'K\xi/\sqrt{a'Wa}$ on the positive-variance event)

$$\frac{a'\,\mathrm{vec}(\hat{B} - B)}{\sqrt{a'\,\hat{\Omega}_B\, a/n}} \to_d N(0,1)$$

on the event $\{a'\Omega_B a > 0\}$. This is the multivariate analogue of the $t$-statistic result of PS3 Problem 1 (cf. Lecture 10): stable convergence of a sequence with a consistent positive-definite variance estimator yields a standard-normal $t$-statistic. A standard $(1-\alpha)$ confidence interval for $a'\,\mathrm{vec}(B)$:

$$a'\,\mathrm{vec}(\hat{B}) \pm z_{1-\alpha/2}\sqrt{a'\,\hat{\Omega}_B\, a\,/\,n}.$$

For an individual entry $B_{ij}$ (the $i$-th coefficient on the $j$-th shock), the standard error simplifies to $\sqrt{[\hat{M}_n\hat{Q}_{xx}^{-1}\hat{M}_n']_{ii}/n}$, since the $(j,j)$ block of $\hat{\Omega}_B$ is $\hat{M}_n\hat{Q}_{xx}^{-1}\hat{M}_n'$ regardless of $j$. Joint inference on multiple linear contrasts follows by the continuous mapping theorem applied to a Wald statistic:

$$T_n = n\big(\Lambda\,\mathrm{vec}(\hat{B}) - r_0\big)'\big(\Lambda\,\hat{\Omega}_B\,\Lambda'\big)^{-1}\big(\Lambda\,\mathrm{vec}(\hat{B}) - r_0\big) \xrightarrow{d} \chi^2_{\mathrm{rank}(\Lambda)}$$

under $\Lambda\,\mathrm{vec}(B) = r_0$, on the event $\{\Lambda\Omega_B\Lambda' \succ 0\}$.

*Assumptions used:* iid $(X_t, Y_t)$ with $m$ fixed, $n \to \infty$; $\mathbb{E}\|X_t\|^{2+\delta}, \mathbb{E}\|u_t\|^{2+\delta} < \infty$ for some $\delta > 0$ (Lindeberg for $\hat{C}$); conditional homoskedasticity $\mathbb{E}[v_t v_t' \mid X_t] = \Sigma$; $\mathbb{E}[X_t X_t']$ invertible; $V = [v_1', \ldots, v_m']'$ invertible almost surely; for each contrast $a$ considered, $a'\Omega_B a > 0$ a.s. (non-degeneracy of the mixed-normal variance). □